

Sebastian Anderson

Embedded Systems & IoT Firmware Engineer • 4 Years Experience • MERIQ Index: 94% (Mastered)

■ sebastian.anderson855@example.com | ■ +1 (225) 309-2443 | ■ Toronto, Canada

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: ARM Cortex, I2C/SPI/UART, FreeRTOS, Python, Embedded Linux, MQTT, C/C++, PCB Debugging

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *Qualcomm* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core ARM Cortex service, reducing latency by 51%.
- Led a squad of 7 engineers delivering scalable microservices processing 15M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 40%.

Associate Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2022 - 2023)

- Designed and maintained I2C/SPI/UART data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using FreeRTOS saving \$16k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [ARM Cortex, I2C/SPI/UART, FreeRTOS, Python]

Engineered end-to-end production solution leveraging ARM Cortex and I2C/SPI/UART with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [FreeRTOS, Python, Embedded Linux, MQTT]

Architected high-resilience system integrating FreeRTOS with 99.9% uptime.

EDUCATION & CREDENTIALS

B.Tech in Computer Science and Engineering

Georgia Institute of Technology — 2018 - 2022 (Score: 8.8 / 10.0)